

SmartBimPei - User Manual

SmartBimPei is the premier software suite integrated into Revit®, designed to simplify the lives of BIM Managers, Coordinators, Designers, and Company Executives. This manual will guide you through using the plugin's 5 key modules.

Module 1: Smart Clash Detector (Geometric Synthesis Engine)

Forget time-consuming clash detection processes. SmartBimPei finds clashes, understands them, and offers concrete solutions in a single click.

 Main Interface of Smart Clash Detector

1. The Configuration Panel (Left)

This panel allows you to set up your search "surgically".

- **Selection Model A & Model B:** Precisely choose the models and categories (e.g., MEP Ducts against Architectural Walls) to cross-check, avoiding unnecessary full-model analysis.
- **Tolerance (mm):** Define the acceptable penetration distance to avoid "false positives" (e.g., insulation slightly touching a wall).
- **"Min. Size" Filters:** Enter a minimum dimension below which the conflict is ignored. Drastically reduces calculation time!
- **Auto-Opening Margin:** Specifies the offset (e.g., 50mm) added around the duct when generating openings to anticipate insulation clearance.

2. The Synthesis Table (Center)

- **Smart Grouping:** The plugin automatically groups very close clashes under a single ID, reducing visual fatigue for the coordinator.
- **"Opening Dimension" Column:** Displays in real-time the final dimensions of the hole/opening to be cut. The architectural impact is immediately visible.

3. Resolution Actions (Bottom)

-  **Isolate Clash:** Opens a centered 3D view. The plugin automatically applies a clear graphical template: a transparent grey environment with the clashing elements opaque and highly contrasted. Instant understanding guaranteed.

 Isolated 3D Visual Rendering

-  **Generate Auto-Opening(s)**: In one click, the plugin places "void" families at the exact locations. The tool handles the creation of rectangular and circular voids in a fraction of a second, a task that would have taken you hours manually.



4. Exports (Communication)

-  **BCF Export**: Generates a `.bcfzip` file incorporating perfect cameras (the famous "Magic Zoom" that prevents having your head stuck in a wall), statuses, and comments for your BIMcollab or BIM Track platforms.



-  **PDF & CSV Export**: Perfect for quickly extracting a visual or Excel report for the coordination meeting.



Module 2: BIM Health Dashboard

The ultimate tool for Management and BIM Managers. This module acts as the control tower for your project. At a glance, you get:

- **The Global Health Score** out of 100 for your model.
- The total number of **Unresolved Clashes**.
- The number of **MEP Duplicates** detected.
- The global **Financial Estimate** of the networks (connected to the QTO module).

***Added Value:** Ideal for evaluating the quality of a received model or for showcasing the cleanup work accomplished by the BIM team.*



Module 3: Duplicate Detector (Model Cleaner)

This module launches an ultra-fast analysis to find all perfectly overlapping elements (often copied by mistake).

- Select the categories to scan (Ducts, Pipes, etc.).
- The table lists the exact duplicates.
- Click **Delete Selection** to purge the model.
- **The goal?** Prevent major financial errors during material ordering.

Module 4: BIM Data Audit & Compliance (Quality Control & CMMS)

This is your Data Police. In one click, it validates that all information required for the As-Built (Handover) and CMMS is present.

1. Choose your Trade in the drop-down menu:
 - **Architecture (Interiors & Openings)**
 - **Structure**
 - **Plumbing**
 - **HVAC**
 - **Electrical**
 2. Run the audit.
 3. The plugin will intelligently look for the required information (Fire Rating, Mark, Manufacturer, Model, Room/Space location...).
- (Tip: The plugin is smart enough to look for the data in the object itself, and if it's not there, it will automatically check the Type Properties!)*
4. The table lists the non-compliances, with an explanatory comment on the business impact (e.g., "Without a Manufacturer, impossible to order the spare part"). Double-click to isolate the faulty object and correct its data.



Module 5: Métré Express (Quantities & Pricing)

A tool designed for construction economists and estimating assistance.

- Choose a network (e.g., Circular Duct).
- The table instantly displays the real lengths, areas, and diameters extracted from the 3D model.
- Enter a Unit Price in the dedicated column.
- The total price updates live. You can then export your estimate in Excel format.



This manual is designed to support team upskilling. SmartBimPeï is not just a tool; it's a major competitive advantage for your BIM processes!